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Quadratic residues

Level 5 Advanced · Number theory

Legendre symbols, Euler's criterion, the laws for -1 and 2, quadratic reciprocity, primes dividing quadratic forms and sums of two squares.

By the end you can: You will be able to evaluate Legendre symbols with Euler's criterion and quadratic reciprocity.; You will be able to determine which primes divide values of a quadratic, and use this to rule out solutions of equations.; You will be able to use the two squares theorem and character sums in olympiad proofs..

Lesson

Let n be odd and let p be a prime factor of 2^n+1. Which residues can p have modulo 8? Orders (L4-NT1) only tell us that $\operatorname{ord}_p(2)$ divides $2n$ but not n, and that says nothing about p modulo 8. Instead, multiply $2^n \equiv -1$ by 2. Since $n+1$ is even,

$$-2 \equiv 2^{n+1} = \left(2^{(n+1)/2}\right)^2 \pmod p,$$

so -2 is a square modulo p. The question is now: for which primes is -2 a square? This unit answers that question with any integer in place of -2. Here the answer (Theorem 4 below) is $p \equiv 1$ or $3 \pmod 8$. For example $2^5+1 = 3\cdot 11$ and $2^7+1 = 3\cdot 43$, and 3, 11, 43 are all 3 modulo 8.

Throughout, p denotes an odd prime.

Residues and the Legendre symbol

An integer a with $p \nmid a$ is a **quadratic residue** modulo p if $x^2 \equiv a \pmod p$ has a solution, and a **non-residue** otherwise. The **Legendre symbol** $\left(\frac{a}{p}\right)$ is 1 for a residue, -1 for a non-residue and 0 if $p \mid a$. With this convention the congruence $x^2 \equiv a$ has exactly $1+\left(\frac{a}{p}\right)$ solutions modulo p for every a, since a solution $x \not\equiv 0$ comes with its partner $-x \not\equiv x$.

Theorem 1. Exactly $\frac{p-1}{2}$ of the numbers $1, 2, \ldots, p-1$ are quadratic residues, namely the residues of $1^2, 2^2, \ldots, \left(\frac{p-1}{2}\right)^2$.

Proof. $x^2 \equiv y^2$ if and only if $p \mid (x-y)(x+y)$, that is, $x \equiv \pm y$. So x and $p-x$ give the same square and the listed squares are distinct. □

Theorem 2 (Euler's criterion). If $p \nmid a$ then $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod p$. Consequently $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$ for all a, b.

Proof. Let $t = a^{(p-1)/2}$. By Fermat $t^2 \equiv 1$, so $t \equiv \pm 1$. If $a \equiv b^2$ then $t \equiv b^{p-1} \equiv 1$. Hence the $\frac{p-1}{2}$ residues are all roots of $x^{(p-1)/2} - 1$, which by Lagrange's theorem (L4-NT1) has at most $\frac{p-1}{2}$ roots modulo p. So no non-residue is a root, and $t \equiv -1$ for a non-residue. For the second statement, if $p \mid ab$ both sides are 0; otherwise both sides are congruent to $(ab)^{(p-1)/2}$ and lie in $\{-1, 1\}$, and $p > 2$. □

The multiplicativity is the part I use most: it reduces every symbol to $\left(\frac{-1}{p}\right)$, $\left(\frac{2}{p}\right)$ and $\left(\frac{q}{p}\right)$ for odd primes q. In particular the product of two non-residues is a residue.

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The residues -1 and 2

Theorem 3. $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$, so -1 is a residue if and only if $p \equiv 1 \pmod 4$.

Theorem 4. $\left(\frac{2}{p}\right) = 1$ if and only if $p \equiv \pm 1 \pmod 8$. Hence $\left(\frac{-2}{p}\right) = 1$ if and only if $p \equiv 1$ or $3 \pmod 8$.

Theorem 3 is Euler's criterion with $a = -1$. Note that it reproves the fact from L4-NT1 that odd prime factors of $n^2 + 1$ are 1 modulo 4.

Proof of Theorem 4 (compare two expressions for one product). Let $h = \frac{p-1}{2}$ and multiply the even numbers $2, 4, \ldots, 2h$: the product is $2^h\, h!$. Now replace each even number $2j > h$ by $2j - p$, which is negative with odd absolute value $p - 2j \le h$. The even numbers $2j \le h$ are all the even numbers in $[1, h]$, and the values $p - 2j$ are distinct odd numbers in $[1, h]$; there are $h - \lfloor h/2 \rfloor$ of them, which is the number of odd numbers in $[1, h]$. So the absolute values are $1, 2, \ldots, h$, each once, and

$$2^h\, h! \equiv (-1)^m\, h! \pmod p, \qquad m = h - \lfloor h/2 \rfloor.$$

Cancelling $h!$ and using Euler's criterion, $\left(\frac{2}{p}\right) = (-1)^m$. For $p = 8k+1, 8k+3, 8k+5, 8k+7$ we get $m = 2k,\ 2k+1,\ 2k+1,\ 2k+2$. The statement about -2 follows from multiplicativity and Theorem 3. $\square$

This settles the opening problem. Another application: for $k \ge 2$, every prime factor p of $2^{2^k} + 1$ satisfies $p \equiv 1 \pmod{2^{k+2}}$. Indeed L4-NT1 gives $\mathrm{ord}_p(2) = 2^{k+1}$, so $p \equiv 1 \pmod 8$, so 2 is a residue and $2^{(p-1)/2} \equiv 1$; hence $2^{k+1} \mid \frac{p-1}{2}$. (Euler's factor 641 of $2^{32} + 1$ is $5 \cdot 2^7 + 1$.)

Example 1. Let p be an odd prime such that $q = 4p + 1$ is also prime. Prove that q divides $2^{2p} + 1$, and that 2 has order $q - 1$ modulo q.

Solution.
Note that $2p = \frac{q-1}{2}$, so by Euler's criterion $2^{2p} \equiv \left(\frac{2}{q}\right) \pmod q$. The first claim is therefore the statement that 2 is a non-residue modulo q, and Theorem 4 turns this into a question about q modulo 8. Since p is odd, $4p \equiv 4 \pmod 8$, so $q \equiv 5 \pmod 8$ and $\left(\frac{2}{q}\right) = -1$. Hence $2^{2p} \equiv -1 \pmod q$.
Let $d = \mathrm{ord}_q(2)$. Then $d \mid 4p$ and $d \nmid 2p$, so $d = 4$ or $d = 4p$. If $d = 4$ then $q \mid 2^4 - 1 = 15$, impossible since $q \ge 13$. Hence $d = 4p = q - 1$. For example $p = 3$ gives $2^6 + 1 = 65 = 5 \cdot 13$.

Quadratic reciprocity

Theorem 5 (quadratic reciprocity). For distinct odd primes p and q,

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

That is, $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ unless $p \equiv q \equiv 3 \pmod 4$, in which case $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$.

I will not prove this here. Gauss published several proofs; the standard one at our level extends the argument of Theorem 4 (this is Gauss's lemma) and then counts lattice points in a rectangle. See Ireland and Rosen, *A Classical Introduction to Modern Number Theory*, Chapter 5, or Niven, Zuckerman and Montgomery, *An Introduction to the Theory of Numbers*, Section 3.2. In an olympiad you may quote Theorems 2 to 5 without proof.

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To evaluate $\left(\frac{a}{p}\right)$: factor a, split by multiplicativity, flip each odd prime with reciprocity, reduce, repeat. For example, to see that 5 is a square modulo the prime 2029, note that $\left(\frac{5}{2029}\right) = \left(\frac{2029}{5}\right) = \left(\frac{4}{5}\right) = 1$. Three general formulas, for odd primes p not dividing the top, are worth remembering:

- $\left(\frac{3}{p}\right) = (-1)^{(p-1)/2}\left(\frac{p}{3}\right)$, which is 1 if and only if $p \equiv \pm 1 \pmod{12}$;
- $\left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)$, which is 1 if and only if $p \equiv 1 \pmod{3}$;
- $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$, which is 1 if and only if $p \equiv \pm 1 \pmod{5}$.

For the first, if $p \equiv 1 \pmod 4$ we need $\left(\frac{p}{3}\right) = 1$, so $p \equiv 1 \pmod 3$; if $p \equiv 3 \pmod 4$ we need $\left(\frac{p}{3}\right) = -1$, so $p \equiv 2 \pmod 3$. Combining gives $p \equiv 1$ or $11 \pmod{12}$.

Residues enter problems mostly through the following observation.

Theorem 6. Let $f(n) = an^2 + bn + c$ have integer coefficients, let $D = b^2 - 4ac$, and let p be an odd prime with $p \nmid a$. Then p divides $f(n)$ for some integer n if and only if $\left(\frac{D}{p}\right) \neq -1$.

Proof. $4af(n) = (2an + b)^2 - D$, and $2an + b$ runs through all residues modulo p as n does, because $2a$ is invertible. □

Example 2. Find all primes p for which there is an integer n with $p \mid n^2 + n + 2$.

Solution.
Theorem 6 applies to odd p, so deal with $p = 2$ first: $n^2 + n = n(n+1)$ is even, so $n^2 + n + 2$ is always even and $p = 2$ works. For odd p, $D = 1 - 8 = -7$, so we need $\left(\frac{-7}{p}\right) \neq -1$. The case $p = 7$ works ($n = 3$ gives 14).
For $p \neq 2, 7$, reciprocity gives $\left(\frac{7}{p}\right) = (-1)^{\frac{p-1}{2}\cdot 3}\left(\frac{p}{7}\right) = (-1)^{(p-1)/2}\left(\frac{p}{7}\right)$, and $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.
The two signs cancel: $\left(\frac{-7}{p}\right) = \left(\frac{p}{7}\right)$. The squares modulo 7 are $1, 2, 4$.
So the answer is $p = 7$ together with all primes $p \equiv 1, 2, 4 \pmod 7$. This list already contains $p = 2$, since $2 \equiv 2 \pmod 7$.

Sums of two squares

Theorem 7 (Fermat). Every prime $p \equiv 1 \pmod 4$ is a sum of two squares.
Lemma 8. If $q \equiv 3 \pmod 4$ is prime and $q \mid x^2 + y^2$, then $q \mid x$ and $q \mid y$.
Theorem 9. A positive integer n is a sum of two squares if and only if every prime $q \equiv 3 \pmod 4$ divides n to an even power.

Proof of Theorem 7 (Thue's lemma, a pigeonhole argument). By Theorem 3 there is r with $r^2 \equiv -1 \pmod p$. Let $m = \lfloor \sqrt{p} \rfloor$; then $m < \sqrt{p}$ since p is not a square. There are $(m+1)^2 > p$ pairs (a, b) with $0 \le a, b \le m$, so two different pairs give the same residue of $a + br$, say $a + br \equiv a' + b'r$. Put $u = a - a'$ and $v = b' - b$. Then $u \equiv rv$, so $u^2 \equiv r^2v^2 \equiv -v^2$ and $p \mid u^2 + v^2$. Also $(u, v) \neq (0, 0)$ and $|u|, |v| \le m < \sqrt{p}$, so $0 < u^2 + v^2 < 2p$. Hence $u^2 + v^2 = p$. □

Proof of Lemma 8. If $q \nmid y$ then $(xy^{-1})^2 \equiv -1 \pmod q$, contradicting Theorem 3. So $q \mid y$, and then $q \mid x^2$, so $q \mid x$. □

Proof of Theorem 9. If the condition holds, n is a product of factors $2 = 1^2 + 1^2$, primes $p \equiv 1 \pmod 4$ (Theorem 7) and squares $q^2 = q^2 + 0^2$, and the identity $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$ shows that products of sums of two squares are sums of two squares. Conversely suppose $n = x^2 + y^2$

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and $q \equiv 3 \pmod 4$ divides n. By Lemma 8, $q \mid x, y$, so $n/q^2 = (x/q)^2 + (y/q)^2$ is again a sum of two squares, with v_q smaller by 2. Repeating, we reach a sum of two squares with v_q equal to 0 or 1; the value 1 is impossible by the same argument, so $v_q(n)$ is even. □

Example 3. Prove that the equation $y^2 = x^3 + 23$ has no integer solutions.

Solution.
Small moduli first. If x is even then $y^2 \equiv 23 \equiv 7 \pmod 8$, impossible. If $x \equiv 3 \pmod 4$ then $x^3 \equiv 3$ and $y^2 \equiv 26 \equiv 2 \pmod 4$, impossible. But $x \equiv 1 \pmod 4$ is not ruled out modulo 8 ($x = 1$ gives $24 \equiv 0$), so a different idea is needed.
(Lebesgue's trick.) Add a constant that makes one side a sum of two squares and the other a sum of two cubes: $23 + 4 = 27$, so

$$y^2 + 2^2 = x^3 + 27 = (x + 3)(x^2 - 3x + 9).$$

Suppose $x \equiv 1 \pmod 4$. Then $x^2 - 3x + 9 \equiv 1 - 3 + 9 \equiv 3 \pmod 4$, and $x^2 - 3x + 9 = \left(x - \frac{3}{2}\right)^2 + \frac{27}{4} > 0$. A positive integer that is 3 modulo 4 has a prime factor $q \equiv 3 \pmod 4$, since a product of primes that are 1 modulo 4 is 1 modulo 4. Then $q \mid y^2 + 2^2$, and Lemma 8 gives $q \mid 2$, a contradiction. Hence there are no solutions.
The positivity check is needed: the step "a number that is 3 modulo 4 has a prime factor that is 3 modulo 4" is false for negative numbers such as -1.

Counting with the Legendre symbol

By Theorem 1, $\sum_{a=0}^{p-1} \left(\frac{a}{p}\right) = 0$. Sums of the symbol over a quadratic are almost as simple.

Lemma 10. If $p \nmid c$ then $\displaystyle\sum_{x=0}^{p-1} \left(\frac{x^2 - c}{p}\right) = -1.$

Proof. Count the solutions of $x^2 - y^2 \equiv c$ in two ways. Writing $u = x - y$ and $v = x + y$, which is a bijection on pairs since 2 is invertible, we need $uv \equiv c$: u is any of the $p - 1$ nonzero residues and then v is determined, so there are $p - 1$ solutions. On the other hand, for each x there are $1 + \left(\frac{x^2-c}{p}\right)$ values of y. Hence $p + \sum_x \left(\frac{x^2-c}{p}\right) = p - 1$. □

Example 4. Let p be an odd prime. How many n with $1 \le n \le p - 2$ are such that n and $n + 1$ are both quadratic residues modulo p?

Solution.
We count with indicator functions: for $1 \le a \le p - 1$ the quantity $\frac{1}{2}\left(1 + \left(\frac{a}{p}\right)\right)$ is 1 for a residue and 0 otherwise. So the required number is

$$N = \frac{1}{4} \sum_{n=1}^{p-2} \left(1 + \left(\frac{n}{p}\right)\right)\left(1 + \left(\frac{n+1}{p}\right)\right).$$

Expand into four sums. The first is $p - 2$. The second is $\sum_{n=1}^{p-2} \left(\frac{n}{p}\right) = 0 - \left(\frac{p-1}{p}\right) = -\left(\frac{-1}{p}\right)$.
The third is $\sum_{m=2}^{p-1} \left(\frac{m}{p}\right) = -\left(\frac{1}{p}\right) = -1$.
The fourth is $\sum_{n=1}^{p-2} \left(\frac{n(n+1)}{p}\right)$. The terms $n = 0$ and $n = p-1$ would be 0, so we may sum over all residues n. Since $4n(n + 1) = (2n + 1)^2 - 1$ and $\left(\frac{4}{p}\right) = 1$, each term equals $\left(\frac{(2n+1)^2-1}{p}\right)$, and $2n + 1$ runs over all residues, so Lemma 10 gives -1.

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Therefore $N = \frac{1}{4}\left(p - 4 - \left(\frac{-1}{p}\right)\right)$, that is, $\frac{p-5}{4}$ if $p \equiv 1 \pmod 4$ and $\frac{p-3}{4}$ if $p \equiv 3 \pmod 4$. Check: modulo 13 the residues are $1, 3, 4, 9, 10, 12$ and the pairs are $(3, 4)$ and $(9, 10)$. In particular two consecutive nonzero residues exist for every $p \geq 7$.

Common traps.

- Euler's criterion and Theorems 3 to 5 are for odd primes. Treat $p = 2$, and primes dividing the coefficients, separately; Example 2 shows how.
- Reciprocity flips $\left(\frac{q}{p}\right)$ only for odd primes $q \neq p$. Factor the top first: $\left(\frac{6}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{3}{p}\right)$, and 2 uses Theorem 4.
- For composite moduli, being a square modulo n means being a square modulo every prime power dividing n. A product of symbols equal to 1 proves nothing: $\left(\frac{2}{3}\right)\left(\frac{2}{5}\right) = 1$, but 2 is not a square modulo 15.
- Lemma 8 says q divides both x and y, not that q cannot divide $x^2 + y^2$: $3 \mid 3^2 + 6^2$.
- "A number $\equiv 3 \pmod 4$ has a prime factor $\equiv 3 \pmod 4$" needs the number to be positive.

How to spot this in a contest

The signs that quadratic residues are relevant, in rough order of frequency:

1. A prime p divides an expression that can be rearranged as "$a \equiv$ square", for instance $p \mid n^2 + k$, $p \mid 4n^2 - 3$, or $p \mid 2^{\text{odd}} + 1$ as in the opening. Translate to $\left(\frac{a}{p}\right) = 1$ and compute with reciprocity; the conclusion is a congruence condition on p.
2. An exponent $\frac{p-1}{2}$ appears anywhere: that is Euler's criterion.
3. An equation such as $y^2 = x^3 + k$: look for a constant that turns it into "sum of two squares = product", then find a factor that is 3 modulo 4, or more generally a prime for which some number is a non-residue.
4. Sums of two squares, or $x^2 + y^2$ divisible by something: Lemma 8 and Theorem 9.
5. Counting residues with a condition attached: indicator functions and Lemma 10.
6. "Infinitely many primes p such that $\left(\frac{a}{p}\right) = 1$": a Euclid-type argument with a quadratic polynomial, see L5-NT2.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆⋆ Find all primes $p > 3$ for which the congruence $x^2 \equiv 6 \pmod p$ has a solution. Give the answer as a list of residues modulo 24.

2. ⋆⋆ Let a be an integer and b an odd integer. Prove that $b^2 + 2$ does not divide $a^2 + 1$. Show by an example that the condition that b is odd cannot be dropped.

3. ⋆⋆⋆ The number 2027 is prime. Find the number of ordered pairs (x, y) of integers with $0 \leq x, y \leq 2026$ such that 2027 divides $x^2 + 5y^2 - 1$.

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4. ⋆⋆⋆ Let p be a prime with $p \equiv 3 \pmod 8$. Prove that $p = x^2 + 2y^2$ for some odd positive integers x and y.

5. ⋆⋆⋆ Let n be a positive integer and let $N = \frac{3^{2^n}+1}{2}$.

   (a) Prove that every prime factor p of N satisfies $p \equiv 1 \pmod{2^{n+1}}$, and that $p \equiv 1 \pmod{2^{n+2}}$ if and only if $p \equiv 1 \pmod 3$.

   (b) Prove that N has a prime factor p with $p \equiv 2^{n+1} + 1 \pmod{2^{n+2}}$.

6. ⋆⋆⋆⋆ Let p be a prime with $p \equiv 3 \pmod 8$, and write $p = x^2 + 2y^2$ with x and y positive integers (practice problem 4 shows that this is possible). Prove that

   (a) x is a quadratic residue modulo p if and only if $x \equiv 1$ or $3 \pmod 8$;

   (b) y is a quadratic residue modulo p if and only if $y \equiv 1 \pmod 4$.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. UKMT BMO1 2020/21 Problem 6 ⋆⋆
2. IMO 1996 Problem 4 ⋆⋆
3. EGMO 2016 Problem 6 ⋆⋆⋆
4. UKMT BMO2 2015/16 Problem 4 ⋆⋆⋆
5. UKMT BMO2 2013/14 Problem 3 ⋆⋆⋆
6. USAMO 2008 Problem 1 ⋆⋆⋆
7. IMO 2008 Problem 3 ⋆⋆⋆⋆
8. USAMO 2020 Problem 3 ⋆⋆⋆⋆
9. USAMO 2026 Problem 6 ⋆⋆⋆⋆

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Hints

1. Use $\left(\frac{6}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{3}{p}\right)$; the two factors depend on p modulo 8 and modulo 12.
2. Work out $b^2 + 2$ modulo 8 for odd b. Which primes can divide $a^2 + 1$?
3. For fixed y, the number of suitable x is $1 + \left(\frac{1-5y^2}{p}\right)$. Pull out a constant so that Lemma 10 applies.
4. -2 is a residue modulo p. Copy the pigeonhole proof of Theorem 7, then use parity and the residue of p modulo 8.
5. For (a), find $\mathrm{ord}_p(3)$, then evaluate $3^{(p-1)/2}$ with Euler's criterion and reciprocity. For (b), look at N modulo 3.
6. For an odd prime $q \mid x$, reduce $p = x^2 + 2y^2$ modulo q to find $\left(\frac{p}{q}\right)$, then flip with reciprocity, remembering that $p \equiv 3 \pmod 4$. Then multiply over the prime factors of x.

Solutions

Problem 1

Answer: $p \equiv 1, 5, 19, 23 \pmod{24}$, that is, $p \equiv \pm 1$ or $\pm 5 \pmod{24}$.

Since $p > 3$ we have $p \nmid 6$, and we need $\left(\frac{6}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{3}{p}\right) = 1$, so the two symbols must be equal. By Theorem 4, $\left(\frac{2}{p}\right) = 1$ exactly when $p \equiv \pm 1 \pmod 8$, and by the formula from the lesson, $\left(\frac{3}{p}\right) = 1$ exactly when $p \equiv \pm 1 \pmod{12}$. Both conditions depend only on p modulo 24, and p is one of $1, 5, 7, 11, 13, 17, 19, 23$ modulo 24.

Going through these: 1 gives $(+,+)$; 5 gives $(-,-)$; 7 gives $(+,-)$; 11 gives $(-,+)$; 13 gives $(-,+)$; 17 gives $(+,-)$; 19 gives $(-,-)$; 23 gives $(+,+)$. The symbols agree for $p \equiv 1, 5, 19, 23 \pmod{24}$, and these are the required primes.

Problem 2

Since b is odd, $b^2 \equiv 1 \pmod 8$, so $b^2 + 2 \equiv 3 \pmod 8$. Thus $b^2 + 2$ is odd, at least 3 and congruent to 3 modulo 4. A product of odd primes that are all 1 modulo 4 is 1 modulo 4, so $b^2 + 2$ has a prime factor $q \equiv 3 \pmod 4$.

Suppose $b^2 + 2 \mid a^2 + 1$. Then $q \mid a^2 + 1$, so $a^2 \equiv -1 \pmod q$, and $q \nmid a$ since $q \nmid 1$. So -1 is a quadratic residue modulo q, which contradicts Theorem 3 as $q \equiv 3 \pmod 4$. Hence $b^2 + 2 \nmid a^2 + 1$.

For even b the statement can fail. Take $b = 12$: then $b^2 + 2 = 146 = 2 \cdot 73$, and $73 \equiv 1 \pmod 4$. Indeed $27^2 + 1 = 730 = 146 \cdot 5$, so $a = 27$ works. (For $b = 2, 4, 6, 8, 10$ the number $b^2 + 2$ is $6, 18, 38, 66, 102$, and each has a prime factor 3 or 19, which is 3 modulo 4, so 12 is the smallest positive even example.) $\square$

Problem 3

Answer: 2026

Write $p = 2027$. For fixed y the number of x with $x^2 \equiv 1 - 5y^2$ is $1 + \left(\frac{1-5y^2}{p}\right)$, so the number of

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pairs is

$$N = p + \sum_{y=0}^{p-1} \left(\frac{1 - 5y^2}{p} \right).$$

Let c be the inverse of 5 modulo p. Then $1 - 5y^2 \equiv -5(y^2 - c)$, and by multiplicativity each term is $\left(\frac{-5}{p}\right)\left(\frac{y^2-c}{p}\right)$. Since $p \nmid c$, Lemma 10 gives $\sum_y \left(\frac{y^2-c}{p}\right) = -1$, so $N = p - \left(\frac{-5}{p}\right)$.

Now $p = 2027 \equiv 3 \pmod 4$, so $\left(\frac{-1}{p}\right) = -1$, and by reciprocity $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = \left(\frac{2}{5}\right) = -1$, as $2027 \equiv 2 \pmod 5$ and the squares modulo 5 are 1 and 4. Hence $\left(\frac{-5}{p}\right) = (-1)(-1) = 1$ and $N = 2027 - 1 = 2026$.

Problem 4

By Theorem 4, $\left(\frac{-2}{p}\right) = 1$, so there is r with $r^2 \equiv -2 \pmod p$. Let $m = \lfloor \sqrt{p} \rfloor$, so $m < \sqrt{p}$ as p is not a square. Among the $(m+1)^2 > p$ pairs (a, b) with $0 \le a, b \le m$, two different pairs give the same residue of $a + br$, say $a + br \equiv a' + b'r$. Put $u = a - a'$ and $v = b' - b$. Then $u \equiv rv$, so $u^2 \equiv -2v^2$ and $p \mid u^2 + 2v^2$. Also $(u, v) \neq (0, 0)$ and $|u|, |v| < \sqrt{p}$, so $0 < u^2 + 2v^2 < 3p$. Hence $u^2 + 2v^2$ equals p or $2p$.

If $u^2 + 2v^2 = 2p$ then u is even, $u = 2w$, and dividing by 2 gives $v^2 + 2w^2 = p$. So in both cases $p = x^2 + 2y^2$ for some integers x, y, which we may take to be non-negative. Since p is odd, x is odd. Also $y \neq 0$, since p is not a square. If y were even then $2y^2 \equiv 0 \pmod 8$ and $p \equiv x^2 \equiv 1 \pmod 8$, contradicting $p \equiv 3 \pmod 8$. Hence x and y are both odd and positive. □

Remark. For example $3 = 1 + 2$, $11 = 9 + 2$, $19 = 1 + 18$, $43 = 25 + 18$.

Problem 5

(a) Since 3^{2^n} is odd, N is an integer and $2N = 3^{2^n} + 1$. As $n \ge 1$, $3^{2^n} = 9^{2^{n-1}} \equiv 1 \pmod 8$, so $2N \equiv 2 \pmod 8$ and N is odd. Also $2N \equiv 1 \pmod 3$, so $3 \nmid N$. Let p be a prime factor of N; then $p \neq 2, 3$.

From $3^{2^n} \equiv -1 \pmod p$ we get $3^{2^{n+1}} \equiv 1$ and $3^{2^n} \not\equiv 1$, as $p \neq 2$. So $\mathrm{ord}_p(3)$ divides 2^{n+1} but not 2^n, that is, $\mathrm{ord}_p(3) = 2^{n+1}$. By Fermat, $2^{n+1} \mid p - 1$. In particular $p \equiv 1 \pmod 4$.

By Euler's criterion $3^{(p-1)/2} \equiv \left(\frac{3}{p}\right) \pmod p$, and since $p \equiv 1 \pmod 4$, reciprocity gives $\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)$. If $p \equiv 1 \pmod 3$, then $3^{(p-1)/2} \equiv 1$, so the order 2^{n+1} divides $\frac{p-1}{2}$, that is, $2^{n+2} \mid p - 1$. If $p \equiv 2 \pmod 3$, then $3^{(p-1)/2} \equiv -1 \not\equiv 1$, so $2^{n+1} \nmid \frac{p-1}{2}$ and $2^{n+2} \nmid p - 1$. This proves both directions.

(b) Since $2N \equiv 1 \pmod 3$, we have $N \equiv 2 \pmod 3$. Every prime factor of N is 1 or 2 modulo 3, and if all were 1 modulo 3 then N would be too. So N has a prime factor $p \equiv 2 \pmod 3$. By (a), $2^{n+1} \mid p - 1$ and $2^{n+2} \nmid p - 1$, so $\frac{p-1}{2^{n+1}}$ is odd, which means $p \equiv 2^{n+1} + 1 \pmod{2^{n+2}}$. □

Remark. Part (b) can also be proved by LTE (L4-NT1): $v_2\left(3^{2^n} - 1\right) = n + 2$, so $N - 1 = \frac{3^{2^n}-1}{2}$ is 2^{n+1} times an odd number, while a product of numbers that are 1 modulo 2^{n+2} is 1 modulo 2^{n+2}. Part (a) needs reciprocity. For $n = 3$, $N = 3281 = 17 \cdot 193$, with $17 = 2^4 + 1$ and $193 = 3 \cdot 2^6 + 1$.

Problem 6

Since p is odd, x is odd. If y were even then $8 \mid 2y^2$ and $p \equiv x^2 \equiv 1 \pmod 8$, which is false; so y is odd. A common prime factor r of x and y would give $r^2 \mid p$, so $\gcd(x, y) = 1$. Also $x, y < \sqrt{p} < p$, so $p \nmid x$ and $p \nmid y$. Since $p \equiv 3 \pmod 4$, reciprocity for an odd prime $q \neq p$ reads $\left(\frac{q}{p}\right) = (-1)^{(q-1)/2}\left(\frac{p}{q}\right) = \left(\frac{-1}{q}\right)\left(\frac{p}{q}\right)$, using Theorem 3.

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Step 1. Let q be a prime factor of x. Then q is odd, $q \neq p$ and $q \nmid y$. Reducing modulo q, $p \equiv 2y^2$, so $\left(\frac{p}{q}\right) = \left(\frac{2}{q}\right)\left(\frac{y}{q}\right)^2 = \left(\frac{2}{q}\right)$. Hence $\left(\frac{q}{p}\right) = \left(\frac{-1}{q}\right)\left(\frac{2}{q}\right) = \left(\frac{-2}{q}\right)$, which by Theorem 4 is 1 if $q \equiv 1, 3 \pmod 8$ and -1 if $q \equiv 5, 7 \pmod 8$.

Step 2. Let q be a prime factor of y. Then q is odd, $q \neq p$ and $q \nmid x$. Reducing modulo q, $p \equiv x^2$, so $\left(\frac{p}{q}\right) = 1$ and $\left(\frac{q}{p}\right) = \left(\frac{-1}{q}\right)$, which is 1 if $q \equiv 1 \pmod 4$ and -1 if $q \equiv 3 \pmod 4$.

Step 3. For odd m let $\alpha(m) = 1$ if $m \equiv 1, 3 \pmod 8$ and $\alpha(m) = -1$ if $m \equiv 5, 7 \pmod 8$, and let $\beta(m) = (-1)^{(m-1)/2}$. Both are multiplicative on odd integers. For α this is a finite check of the products of $1, 3, 5, 7$ modulo 8: $3 \cdot 3 \equiv 1$, $3 \cdot 5 \equiv 7$, $3 \cdot 7 \equiv 5$, $5 \cdot 5 \equiv 1$, $5 \cdot 7 \equiv 3$, $7 \cdot 7 \equiv 1$. For β, write $mm' - 1 = (m-1)(m'-1) + (m-1) + (m'-1)$; the first term is divisible by 4, so $\frac{mm'-1}{2} \equiv \frac{m-1}{2} + \frac{m'-1}{2} \pmod 2$.

Write $x = q_1 q_2 \cdots q_s$ as a product of (not necessarily distinct) primes, with $s = 0$ if $x = 1$. By multiplicativity of the Legendre symbol and Step 1,

$$\left(\frac{x}{p}\right) = \prod_{i=1}^{s} \left(\frac{q_i}{p}\right) = \prod_{i=1}^{s} \alpha(q_i) = \alpha(x).$$

Since $p \nmid x$, x is a residue if and only if $\alpha(x) = 1$, that is, if and only if $x \equiv 1$ or $3 \pmod 8$. In the same way Step 2 gives $\left(\frac{y}{p}\right) = \beta(y)$, so y is a residue if and only if $y \equiv 1 \pmod 4$. $\square$

Check. $11 = 3^2 + 2 \cdot 1^2$ and $3 \equiv 5^2 \pmod{11}$; $43 = 5^2 + 2 \cdot 3^2$, and both 5 and 3 are non-residues modulo 43; $59 = 3^2 + 2 \cdot 5^2$, and $3 \equiv 11^2$, $5 \equiv 8^2 \pmod{59}$.

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